

6.6 Fatjet top-tagging

In order to identify the hadronically decaying top quark in this analysis, a Graph-Neural-Network-based top-tagger (`particleNet`) is used [71]. This tagger aims at identifying the physical origin of a fatjet by assigning probability-like scores based on the jet properties. For our use case the discrimination between jets from QCD processes versus jets originating from hadronic top decays is mainly used. However, this tagger was trained on AK8 jets and is in this analysis applied to AK15 jets. Therefore, the top-tagging discriminant calculated for the leading AK15 jet is investigated in the following in more detail i.e. its distribution for different kinds of processes is studied.

Figure 21 shows a shape comparison of all processes considered in the hadronic signal region. As expected the distribution peaks towards the right-hand side of the distribution for processes containing real top quarks (monotop signal, $t\bar{t}$, Single- t), whereas the distribution peaks towards the left-hand side for processes with fatjets that can only originate from QCD-initiated processes ($W(\ell\nu) + \text{jets}$, $Z(\ell\ell) + \text{jets}$, $Z(\nu\nu) + \text{jets}$, $\gamma + \text{jets}$). A strong suppression of events with a QCD-initiated leading AK15 jet can therefore be achieved.

However, a significant number of events, in which a real hadronically decaying top quark might be expected, is nevertheless accumulated towards the left-hand side of the distribution.

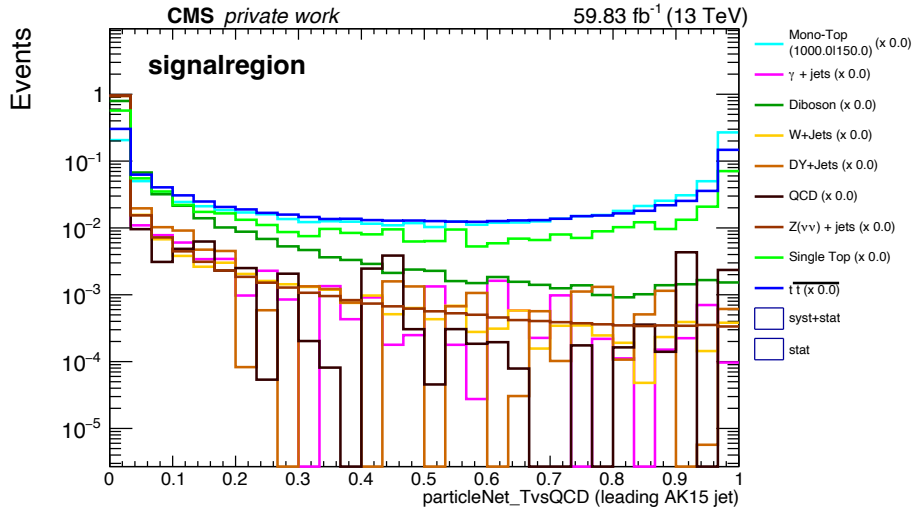


Figure 21: Normalized shape comparison of the `particleNet` top vs. QCD tagger for different processes in the hadronic signal region for the 2018 era.

This feature is studied in more detail later on. It can already be concluded here that a powerful discriminator can be built using this tagger.

6.6.1 Top-tagging efficiency measurement

6.6.1.1 Introduction The decay products of a hadronically decaying top quark can be clustered into a fatjet if the transverse momentum of the top quark is large enough. If this is not the case, not all decay products may be clustered into the fatjet. In addition, fatjets can also occur by pure QCD processes without any heavy resonance involved. Consequently, the AK15 jets used in this analysis are first categorized into different classes according to which extent a hadronically decaying top quark is merged into the AK15 jet, whether it is a purely QCD-initiated AK15 jet, and whether heavy-flavor quarks are merged within the AK15 jets. This is

done to understand better how the tagger responds to different kinds of jets, which is necessary for the derivation of sensible efficiencies in simulation and data as well as corresponding scale factors. The categorization of the AK15 jets is defined as follows:

1. No quark from the decay of a top quark is found within (via distance $\Delta R \leq 1.5$) the AK15 jet and **no** b-flavor is found within the AK15 jet (via ghost-hadron matching).
2. No quark from the decay of a top quark is found within the AK15 jet but b-flavor is found within the AK15 jet.
3. One quark from the decay of a top quark is found within the AK15 jet and **no** b-flavor is found within the AK15 jet.
4. One quark from the decay of a top quark is found within the AK15 jet and b-flavor is found within the AK15 jet.
5. Two quarks from the decay of a top quark are found within the AK15 jet and **no** b-flavor is found within the AK15 jet.
6. Two quarks from the decay of a top quark are found within the AK15 jet and b-flavor is found within the AK15 jet.
7. Three quarks from the decay of a top quark are found within the AK15 jet.
8. More than three quarks from the decay of a top quark are found within the AK15 jet (this can happen if e.g. a top quark-antiquark pair is in an event).

For this study, events from the different analysis regions, see section 7, are used. For the leading AK15 jet, the category is then determined and the matching number, corresponding to the list number above, is filled into a histogram. The resulting distributions are shown in Fig. 22.

It is observed that, as expected, the processes not involving any top quarks i.e. the V + jets processes, have a very large number of QCD-initiated jets without b flavor contained in the jet and a significantly smaller number of QCD-initiated jets containing b flavor. The ratio of the numbers is, also as expected, quite similar for the different V + jets processes. A slightly larger ratio of the QCD-jet contribution with b flavor and without b flavor compared to the γ + jets region and the $W(\ell\nu)$ + jets region is observed in the $Z(\ell\ell)$ + jets region because in the latter no b-tagging criterium is applied to the AK4 jets outside of the leading AK15 jet. This is done to keep more statistics in this region.

On the other hand, the main background process containing genuine top quarks ($t\bar{t}$) and the signal process have contributions mainly in the jet class range ≥ 3 and a small contribution in jet class 1. This is also expected because all sorts of combinations of the products of the hadronic top quark decay might be clustered into the AK15 jet, but it can also happen that e.g. a high-energy QCD radiation might be clustered as an AK15 jet. It is also reassuring that the jet class with the largest contribution for the $t\bar{t}$ process is the one representing the clustering of the three quarks of the hadronic decay of the top quark.

A perfect calibration of the top-tagger would consider all relevant jet classes in the determination of efficiencies and scale factors. However, this is not possible since for each jet class, a dedicated phase space enriched in this kind of jet would need to be defined to be able to determine efficiencies also in data and therefore also to calculate scale factors. Because of this, several of the aforementioned jet classes are summarized in a lesser number of jet classes for which control regions, in which efficiencies can be calculated in simulation and data, can be found.

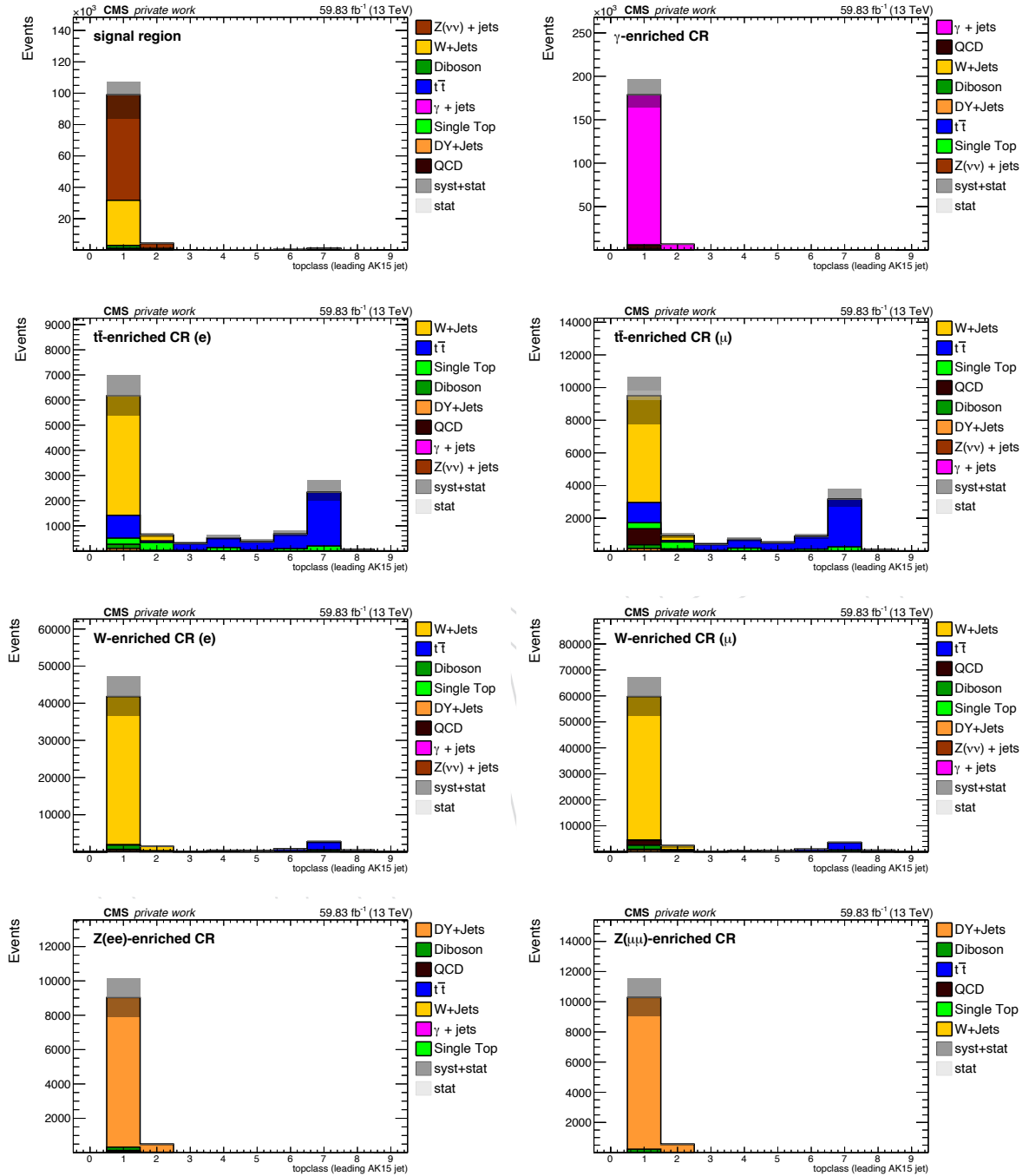


Figure 22: Jet class of the leading AK15 jet according to the categorization scheme of section 6.6.1 without any requirement on the top-tagger value i.e. only the event preselection is applied.

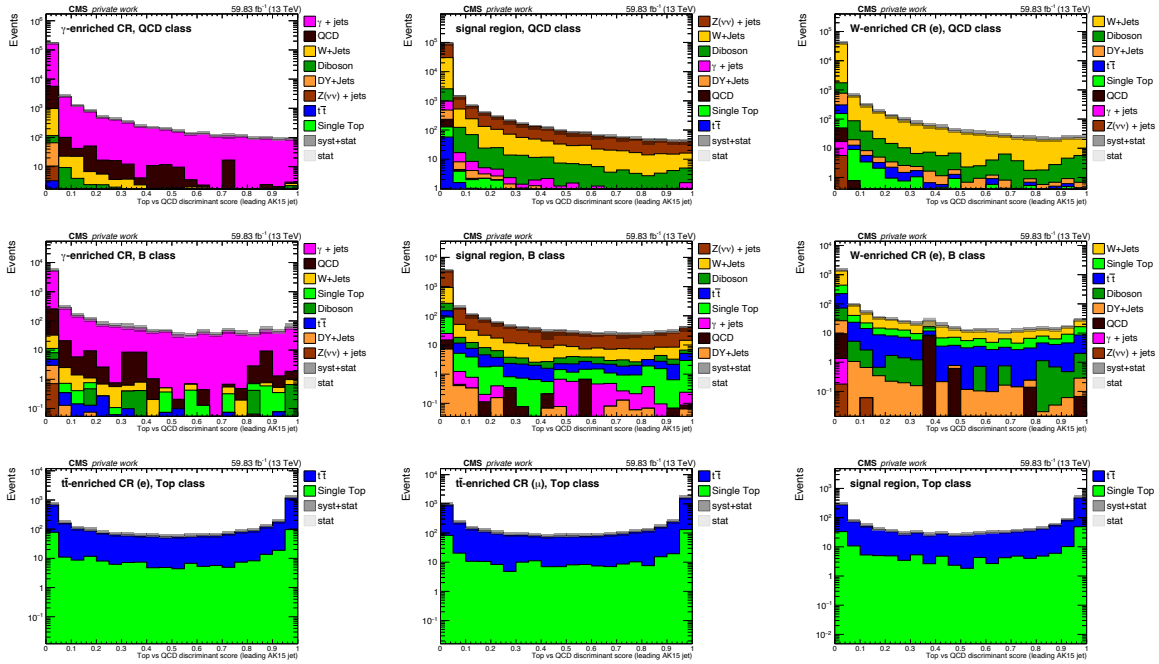


Figure 23: Top-tagging discriminant distributions in several analysis regions if the jet is a QCD-jet (top row), B-jet (middle row), Top-jet (bottom row).

6.6.1.2 Working point

Based on the findings of the previous section, three aggregated AK15 jet classes are defined:

- Top: Classes 5, 6, 7 and 8 are combined into the **Top-jet** class.
- B: Classes 2 and 4 are combined into the **B-jet** class.
- QCD: Classes 1 and 3 are combined into the **QCD-jet** class.

A dedicated B-jet class is considered, because it is expected that a top-tagger reacts to b flavor with a higher/different efficiency than to purely light flavor jets. How the three defined classes affect the top-tagging discriminant is shown in Fig. 23. It is qualitatively observed that the different jet classes, as naively expected, show different distributions of the top-tagging discriminant. The QCD-jet class has a steeply falling distribution, however the slope of the decrease becomes smaller the higher the value of the top-tagging discriminant. Moreover, the B-jet class has a steeply falling distribution as well, however the slope is not as large as in the QCD-jet class and also the distribution starts to slowly rise again for high values of the top-tagging discriminant. Finally, the distributions of the Top-jet class fall as well at low discriminant values but then start to rise again leading into a strong rise for high discriminant values. The behavior qualitatively seems to be independent of the analysis region as well.

As the next step, a ROC curve is derived to investigate the separation power of the top-tagger between the Top-jet class and the QCD-jet class in the signal region in order to define a working point. The Top-jets are taken from the $t\bar{t}$ process whereas the QCD-jets are taken from the $Z(\nu\nu) + \text{jets}$ process, which is the main background process in the signal region. The corresponding ROC curve is shown on the left-hand side of Fig. 24. It is observed that a very strong separation power between Top-jets and QCD-jets is achieved. The background rejection increases very fast and reaches a background rejection of more than 95% with a signal efficiency of over 80% at a working point of 0.05. As working point for this analysis, however, a discriminant value of 0.26 is chosen. With this working point a background efficiency (rejec-

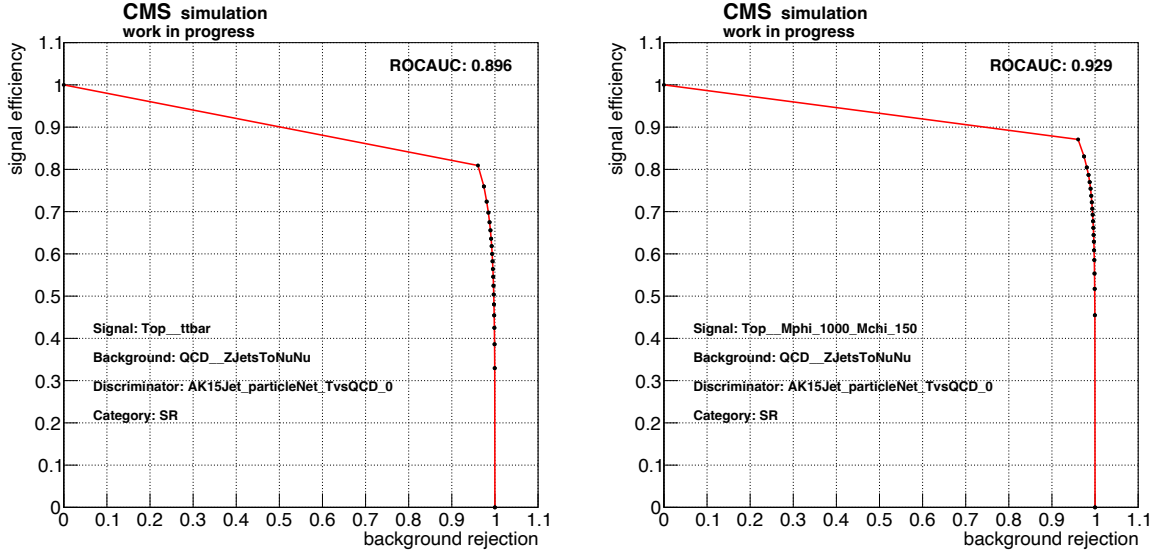


Figure 24: On the left-hand side, the ROC curve obtained from the top-tagging discriminant used to distinguish Top-jets from $t\bar{t}$ and QCD-jets from $Z(\nu\nu) + \text{jets}$ in the signal region is shown. On the right-hand side, a similar ROC curve is shown, however the Top-jets are obtained from a signal sample with mediator mass 1000 GeV and Dark Matter mass 150 GeV. The given ROCAUC values are underestimated because they are calculated based on the red interpolated curve.

tion) of $\approx 1\%$ (99%) is obtained together with a signal efficiency of $\approx 67\%$. With this approach a high fraction of genuine Top-jets can be retained whereas a very large fraction of QCD-jets can be rejected. Going even higher with the working point would lead to a severe loss in signal efficiency with only small gains in background rejection.

The efficiencies also depend on the transverse momentum of the AK15 jets. If Top-jets from a signal sample are chosen for this comparison an even stronger separation is obtained, see right-hand side of Fig. 24. This is due to the on-average higher-momenta AK15 jets, which have a higher efficiency to be tagged, as will be shown later.

6.6.1.3 Efficiencies The efficiencies can be determined from some of the control regions shown above. As mentioned earlier, this is done in bins of AK15 jet p_T . The top-tagger calibration (calculation of efficiencies and scale factors) is performed in the following four bins of the leading AK15 jet p_T :

- Bin 1: $250 \text{ GeV} \leq p_T < 325 \text{ GeV}$
- Bin 2: $325 \text{ GeV} \leq p_T < 400 \text{ GeV}$
- Bin 3: $400 \text{ GeV} \leq p_T < 600 \text{ GeV}$
- Bin 4: $600 \text{ GeV} \leq p_T$

Specifically, the QCD-jet misidentification fraction is determined from the $\gamma + \text{jets}$ control region, see Fig. 25 for the different jet class contributions in this region. This region comprises of a large QCD-jet part, a smaller B-jet part and a negligible Top-jet part. This allows to determine the selection efficiency of the QCD-jet class in simulation based on a large number of events/jets.

The Top-jet identification efficiency as a function of the jet p_T is determined from the combina-

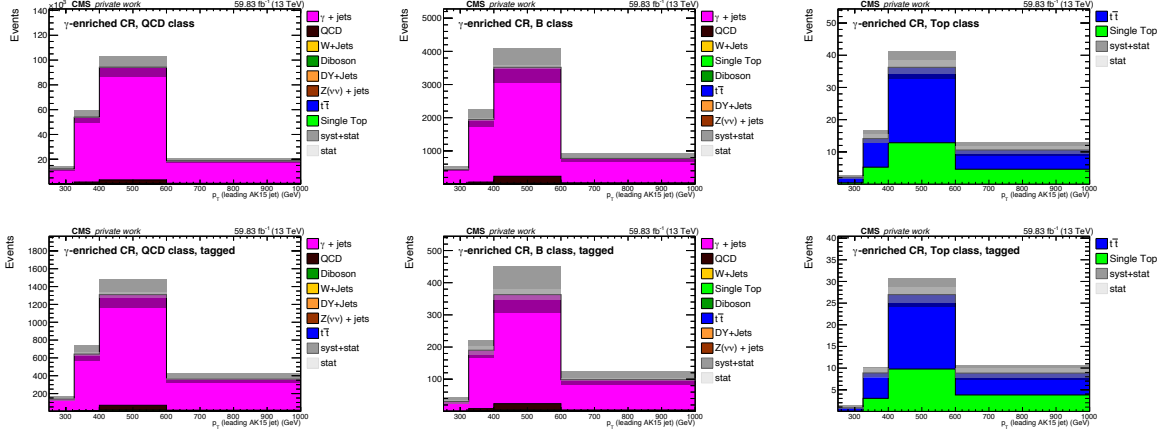


Figure 25: Leading AK15 jet p_T distribution in the γ + jets control region if the jet is a QCD-jet (left), B-jet (center), Top-jet (right). The top row is without a top-tag and the bottom row with a top-tag.

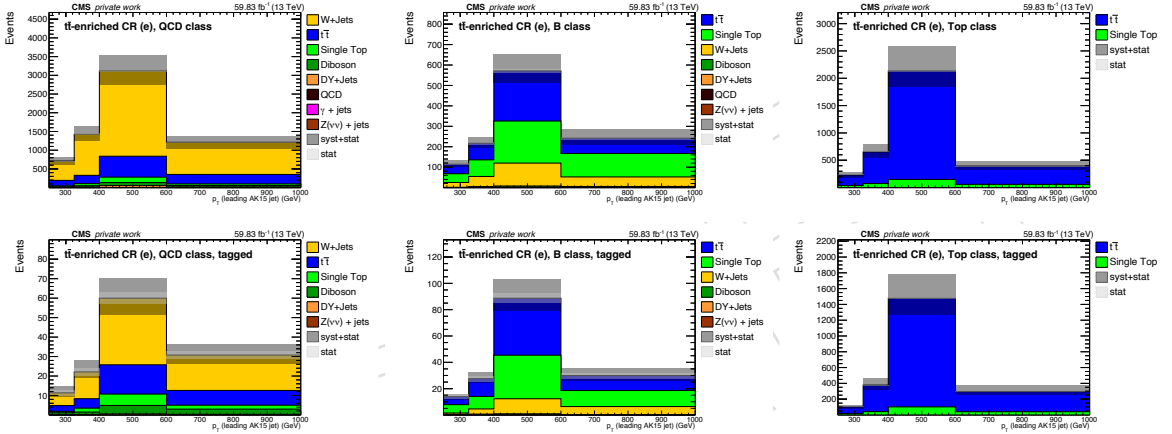


Figure 26: Leading AK15 jet p_T distribution in the $t\bar{t}$ (e) control region if the jet is a QCD-jet (left), B-jet (center), Top-jet (right). The top row is without a top-tag and the bottom row with a top-tag.

tion of the $t\bar{t}$ electron and muon control region, see Fig. 26 and Fig. 27 for their composition.

Since no dedicated region is readily available to determine the B-jet efficiency in data, and therefore a scale factor, a conservative per-bin scale factor of $SF_{B\text{-jet}} = 1 \pm 0.5$ is assumed instead. This scale factor is then considered for and its uncertainty propagated to the determination of the QCD-jet and Top-jet scale factors. Since the B-jet contamination is neither the largest contribution in the γ + jets control region nor in the $t\bar{t}$ control region, this choice should be conservative enough to cover the actual scale factors, but also its influence on the calibration results for the Top-jet and QCD-jet classes is limited (but still considered).

The following procedure is performed in **each** of the aforementioned AK15 jet p_T bins:

QCD-jet calibration The QCD-jet selection efficiency (mistag fraction) of the top-tagging algorithm in simulation is readily determined from the γ + jets simulation in the γ + jets control region by

$$\epsilon_{\text{QCD-jet, MC}} = \frac{N(\gamma + \text{jets, QCD-jet, tagged})}{N(\gamma + \text{jets, QCD-jet})}$$

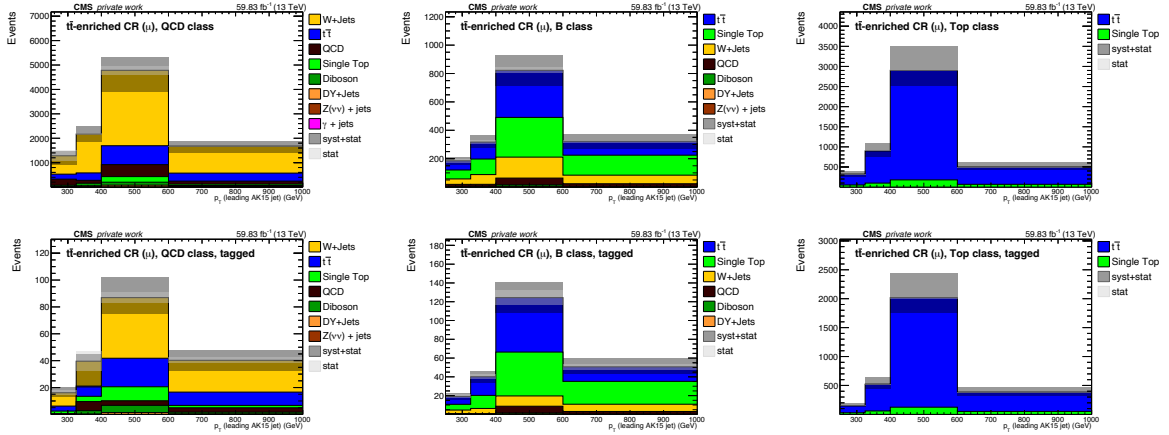


Figure 27: Leading AK15 jet p_T distribution in the $t\bar{t}$ (μ) control region if the jet is a QCD-jet (left), B-jet (center), Top-jet (right). The top row is without a top-tag and the bottom row with a top-tag.

995 with $N(\gamma + \text{jets, QCD-jet, tagged})$, referring to the expected yield of the $\gamma + \text{jets}$ process with the
 996 leading jet belonging to the QCD-jet class and being top-tagged by the top-tagger, as well as
 997 $N(\gamma + \text{jets, QCD-jet})$, being the same expected yield but without the criterion of the jet being
 998 top-tagged.

Before calculating the efficiency in data, the total simulation (in each bin) is scaled by the factor

$$\alpha = \frac{N(\text{Data})}{N(\text{MC})}$$

with $N(\text{Data})$ representing the number of observed events in data in the corresponding AK15 jet p_T bin and $N(\text{MC})$ representing the corresponding total number of expected events independent of the top-tagging status or the jet class. The ratio α is therefore the data/MC ratio before the application of the top-tagger. By doing this scaling, data/MC discrepancies which are observed even before the application of the top-tagging algorithm, can be mitigated such that these discrepancies do not affect the calculation of the efficiencies and scale factors. The corresponding efficiency in data is then calculated by

$$\begin{aligned} \epsilon_{\text{QCD-jet, Data}} &= \frac{N(\text{Data, tagged}) - \alpha \cdot N(\text{MC, B-jet, tagged})}{N(\text{Data}) - \alpha \cdot N(\text{MC, B-jet})} \\ &\approx \frac{N(\text{Data, tagged}) - \alpha \cdot N(\gamma + \text{jets, B-jet, tagged})}{N(\text{Data}) - \alpha \cdot N(\gamma + \text{jets, B-jet})}. \end{aligned}$$

The approximation to exchange the total simulation (MC) with the $\gamma + \text{jets}$ MC is performed because this process is by far the most dominant process in the $\gamma + \text{jets}$ region, see Fig. 25. Using that, by definition,

$$N(\text{Data}) = \alpha \cdot N(\text{MC}) \approx \alpha \cdot (N(\gamma + \text{jets, QCD-jet}) + N(\gamma + \text{jets, B-jet})),$$

and also

$$\epsilon_{\text{QCD-jet, MC}} = \frac{\alpha \cdot N(\gamma + \text{jets, QCD-jet, tagged})}{\alpha \cdot N(\gamma + \text{jets, QCD-jet})},$$

the QCD-jet tag scale factor is then determined via

$$\text{SF}_{\text{QCD-jet}} = \frac{\epsilon_{\text{QCD-jet, Data}}}{\epsilon_{\text{QCD-jet, MC}}} = \frac{N(\text{Data, tagged}) - \alpha \cdot N(\gamma + \text{jets, B-jet, tagged})}{\alpha \cdot N(\gamma + \text{jets, QCD-jet, tagged})}.$$

All uncertainties considered in this analysis are propagated into this method and the resulting uncertainties are added in quadrature to a total uncertainty. Also, the aforementioned uncertainty on the assumed B-jet scale factor is propagated and an additional 50% normalization uncertainty on the γ + jets B-jet simulation is added to account for the difficult modeling of vector boson production in association with heavy flavor jets.

Top-jet calibration The Top-jet selection efficiency (tag efficiency) is calculated in an analogous manner, however the $t\bar{t}$ electron and muon control regions are used in combination. The efficiency in simulation can be calculated by

$$\epsilon_{\text{Top-jet, MC}} = \frac{N(\text{MC, Top-jet, tagged})}{N(\text{MC, Top-jet})}$$

with MC referring to all relevant background processes in the $t\bar{t}$ control regions. Analogously to before, again the data/MC factor α is introduced. Then it follows

$$\epsilon_{\text{Top-jet, Data}} = \frac{N(\text{Data, tagged}) - \alpha \cdot N(\text{MC, B/QCD-jet, tagged})}{N(\text{Data}) - \alpha \cdot N(\text{MC, B/QCD-jet})}$$

with B/QCD-jet representing the combined contribution of both classes. In this expression the aforementioned assumed B-jet scale factors as well as the previously derived QCD-jet scale factors are used to estimate/correct $N(\text{MC, B/QCD-jet, tagged})$. Also the uncertainties of the B-jet scale factors and QCD-jet scale factors are propagated to the uncertainty of the Top-jet efficiencies and scale factors. Again, it holds that

$$N(\text{Data}) = \alpha \cdot N(\text{MC}) = \alpha \cdot (N(\text{MC, Top-jet}) + N(\text{MC, B/QCD-jet})),$$

and

$$\epsilon_{\text{Top-jet, MC}} = \frac{\alpha \cdot N(\text{MC, Top-jet, tagged})}{\alpha \cdot N(\text{MC, Top-jet})}.$$

Using the two previous equations, the Top-jet scale factors are derived by

$$\text{SF}_{\text{Top-jet}} = \frac{\epsilon_{\text{Top-jet, Data}}}{\epsilon_{\text{Top-jet, MC}}} = \frac{N(\text{Data, tagged}) - \alpha \cdot N(\text{MC, B/QCD-jet, tagged})}{\alpha \cdot N(\text{MC, Top-jet, tagged})}.$$

analogously to before. All uncertainties are propagated into this method and the resulting uncertainties are added in quadrature to a total uncertainty for the efficiencies and the scale factors.

Results The determined efficiencies and scale factors for the QCD-jet class and the Top-jet class in the 2018 era are shown in Fig. 28.

First, it is observed that the selection efficiency for the QCD-jet class (QCD-jet mistag fraction) is very different between data and simulation. For simulation it ranges between 1 – 2% depending on the considered p_T bin. Both in data and simulation, the efficiencies increase with increasing transverse momentum of the AK15 jet. Altogether this leads to data/MC scale factors in the range of 2 – 2.5.

The selection efficiency for the Top-jet class (Top-jet tag efficiency) behaves differently. For the Top-jet class the differences between data and simulation are much smaller. The efficiency in simulation is between 45 – 80% again depending on the considered AK15 jet p_T bin. Both efficiencies increase with increasing jet momentum. The corresponding scale factors then range

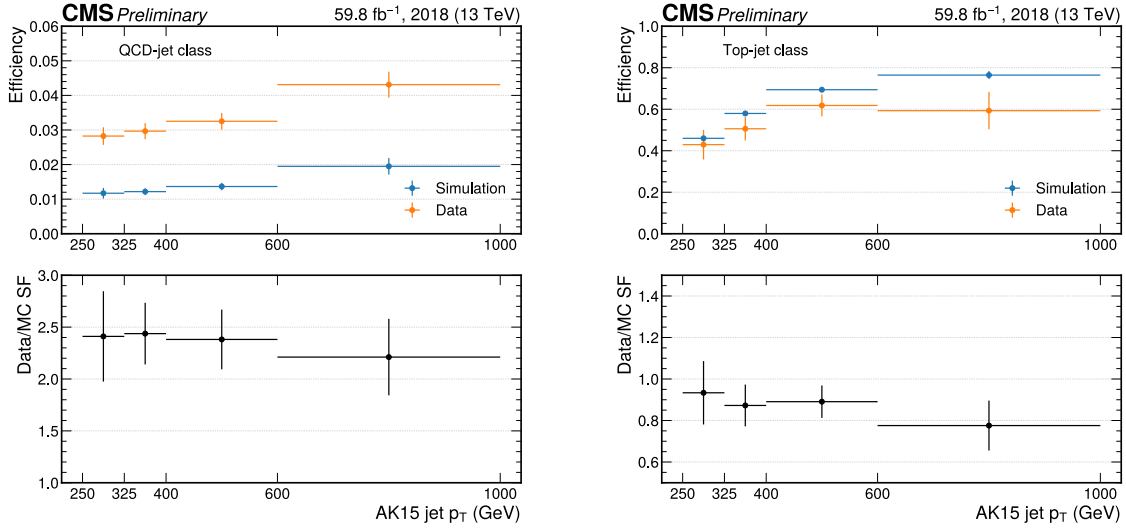


Figure 28: Selection efficiency of the top-tagging algorithm at a working point of 0.26 for the QCD-jet class (left-hand side) and the Top-jet class (right-hand side) in four different p_T bins of the leading AK15 jet p_T .

between 0.8 – 1.0. The efficiencies and scale factors determined for the other eras are shown in Appendix C.1 and confirm the findings shown here.

While we are aware that scale factors in the range of 2 – 2.5 seem odd, these findings however are confirmed when the distributions of the top-tagging discriminant are studied as shown in Appendix D.1.3 ($Z(\ell\ell)$ + jets region), D.1.1 (W + jets region), and D.1.2 ($t\bar{t}$ region). The $Z(\ell\ell)$ + jets region is also heavily enriched in events that have a leading AK15 jet which belongs to the QCD-jet class, as was shown above. In the corresponding top-tagger discriminant distribution, it is observed that the data/MC discrepancies become very large for discriminant values above our working point (0.26). The QCD-jet data/MC scale factors therefore seem compatible with this observation. Furthermore, looking at the top-tagger discriminant distribution in the $t\bar{t}$ region, it is observed that the data/MC discrepancies are much smaller. Given that the Top-jet class is the largest contribution for discriminant values above the working point, this is again compatible to the derived scale factors for the Top-jet class being closer to one. Finally, both of the aforementioned effects are seen in the top-tagger discriminant distribution in the W + jets region. In this region, the W + jets contribution mainly consisting of events with AK15 jets belonging to the QCD-jet class, still has a significant contribution for discriminant values far away from one. In this range the data/MC discrepancies are again large. For discriminant values close to one, where again the Top-jet class events from the $t\bar{t}$ process start to dominate, the data/MC agreement becomes again significantly better.

The calibration is applied in an analogous manner as the AK4 b-jet calibration, i.e. an event weight is calculated based on the MC efficiencies and the scale factors to calculate the event probabilities in simulation and data, cf. section 6.5.

$$w_{\text{top-tag}} = \prod_{j \in \text{tagged}} \left(\frac{\epsilon_{j,\text{Data}}}{\epsilon_{j,\text{Sim}}} \right) \prod_{j \in \text{untagged}} \left(\frac{1 - \epsilon_{j,\text{Data}}}{1 - \epsilon_{j,\text{Sim}}} \right) = \prod_{j \in \text{tagged}} \left(\frac{\epsilon_{j,\text{Sim}} \cdot \text{SF}_j}{\epsilon_{j,\text{Sim}}} \right) \prod_{j \in \text{untagged}} \left(\frac{1 - \epsilon_{j,\text{Sim}} \cdot \text{SF}_j}{1 - \epsilon_{j,\text{Sim}}} \right)$$

The set “tagged” (“untagged”) refers to the AK15 jets that are considered for top-tagging, i.e. only the leading AK15 jet, and which are (not) tagged at the used working point. As explained

Table 31: Relative uncertainties of the most important sources of systematic uncertainty for the QCD-jet mistagging scale factors split into p_T bins of the leading AK15 jet. BJetNorm represents the assumed 50% normalization uncertainty on the B-jet contribution in the γ + jets control region and BJetMistag references the assumed 50% uncertainty on the B-jet mistagging scale factors. The total relative uncertainty includes not only the most important sources of uncertainty but all sources.

Source	250-325 GeV	325-400 GeV	400-600 GeV	600-Inf GeV
statistics	15.1 %	6.2 %	4.1 %	6.4 %
jesRelativeBal	3.9 %	1.0 %	0.3 %	0.0 %
jesRelativeSample 2018	5.1 %	1.1 %	0.8 %	0.0 %
factorization scale γ + jets	3.0 %	1.7 %	2.0 %	1.1 %
renormalization scale γ + jets	0.7 %	2.8 %	2.1 %	1.3 %
initial-state parton shower	1.3 %	4.0 %	3.3 %	2.2 %
final-state parton shower	0.7 %	5.0 %	7.8 %	13.3 %
BJetMistag	4.6 %	5.8 %	5.7 %	6.1 %
BJetNorm	2.7 %	3.9 %	3.8 %	3.9 %
total	18.1 %	12.2 %	12.1 %	16.7 %

previously, the efficiencies and scale factors depend on the AK15 jet class (Top-jet, B-jet, QCD-jet) and the transverse momentum.

To conclude this section, the top-tagging discriminant is compared with and without applying the calibration derived in this section. However, since no shape calibration but a fixed working point calibration was derived, the top-tagging discriminant is shown as a histogram with two bins. The first bin ranges from zero to the working point (0.26) and the second bin from the working point to 1. By this representation, it can be checked whether the calibration works as expected. For the 2018 era, the comparison is given in Fig. 29 and 30. Please note the different scale of the y -axis in the ratio pad comparing the plots on the left-hand and right-hand side.

As can be observed, the calibration procedure works quite well. It corrects data/MC discrepancies in the bins ranging from the working point to 1 (later these phase spaces correspond to the regions called "pass"). Especially in the $Z(\ell\ell)$ + jets and γ + jets control regions, the strong data/MC discrepancy is corrected. Note that the 5 – 10% overprediction in the γ + jets control region is the situation before applying any top-tagging, so this is recovered as well. Moreover, also in the W + jets and $t\bar{t}$ control regions, the calibration consistently improves the modeling. The same comparisons for the other eras are given in Appendix C.2.

A comparison of the controlplots showing the hadronic recoil when using top-tagging with and without the derived calibration scale factors is performed in section 8.

Uncertainties of the top-tagging calibration As was mentioned earlier, all uncertainties this analysis considers are propagated to the determination of the efficiencies and scale factors. In the two tables Tab. 31 and Tab. 32 the most important systematic uncertainties and their relative size compared to the scale factors themselves are provided. The total relative uncertainty includes not only the most important sources of uncertainty but all sources.

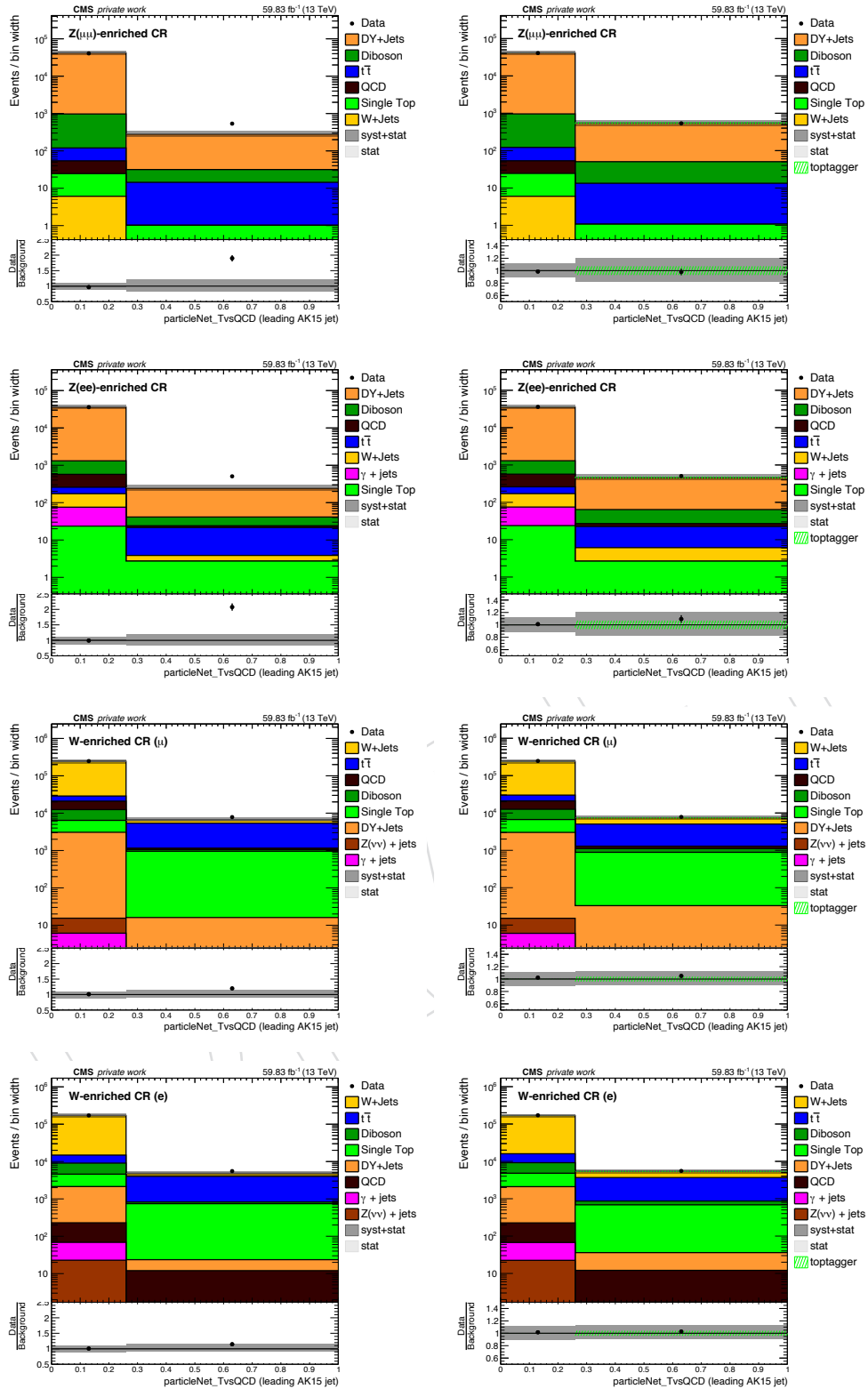


Figure 29: Data/MC comparison of the top-tagger discriminant in the 2018 era. On the left-hand side no top-tagger calibration is applied, on the right-hand side the calibration derived in this analysis is applied. Please note the different scale of the y -axis in the ratio pad comparing the plots on the left-hand and right-hand side.

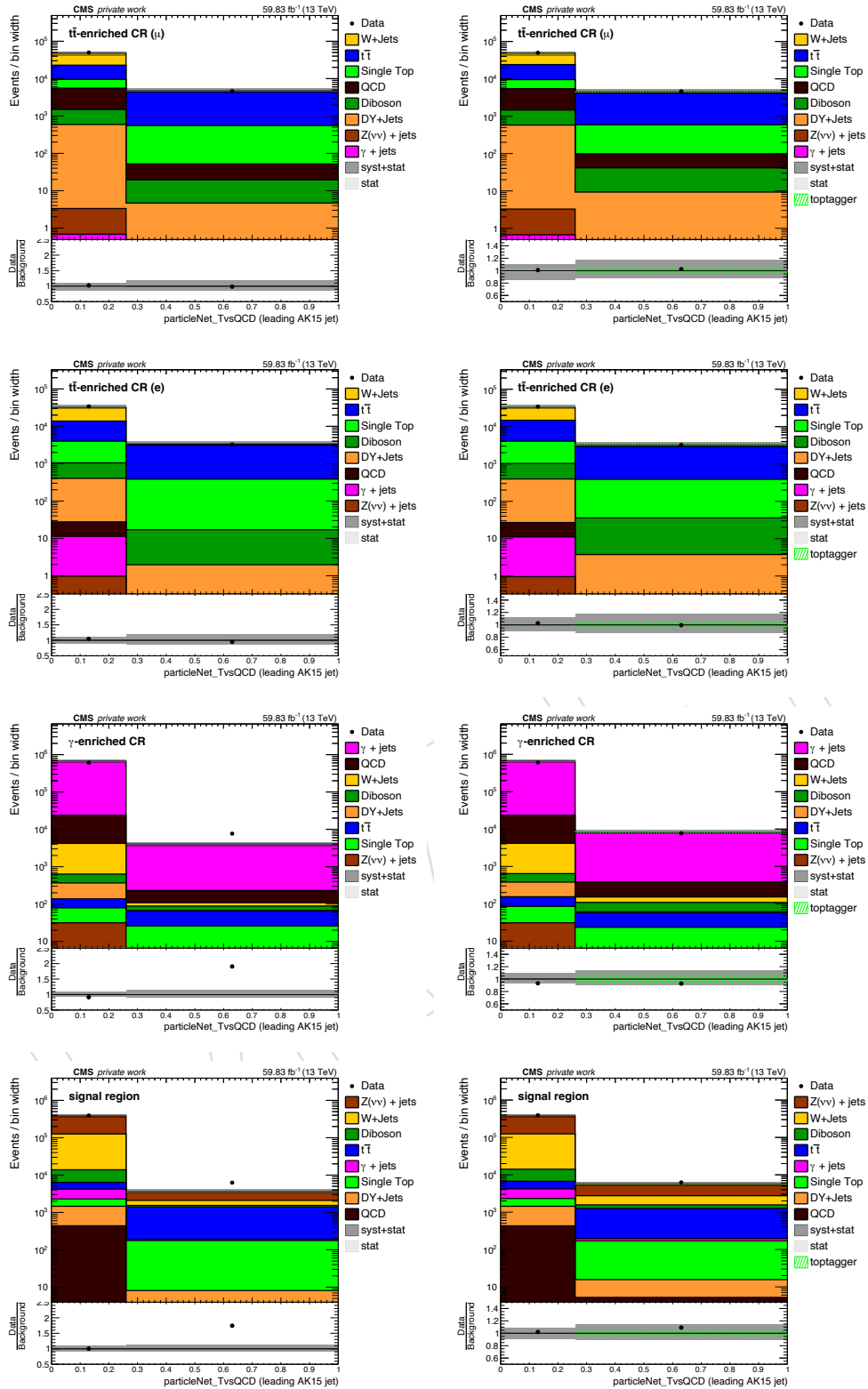


Figure 30: Data/MC comparison of the top-tagger discriminant in the 2018 era. On the left-hand side no top-tagger calibration is applied, on the right-hand side the calibration derived in this analysis is applied. Please note the different scale of the y -axis in the ratio pad comparing the plots on the left-hand and right-hand side.

Table 32: Relative uncertainties of the most important sources of systematic uncertainty for the Top-jet tagging scale factors split into p_T bins of the leading AK15 jet. BJetMistag references the assumed 50% uncertainty on the B-jet mistagging scale factors and QCDMistag the uncertainty on the previously determined QCD-jet mistagging scale factors. The total relative uncertainty includes not only the most important sources of uncertainty but all sources.

Source	250-325 GeV	325-400 GeV	400-600 GeV	600-Inf GeV
statistics	9.6 %	5.4 %	2.4 %	6.7 %
jesFlavorQCD	3.4 %	2.6 %	0.4 %	0.1 %
jesRelativeSample_2018	2.3 %	0.6 %	0.6 %	0.3 %
TopPt_reweighting	5.6 %	6.1 %	6.7 %	9.4 %
renormalization scale $t\bar{t}$	3.9 %	2.2 %	0.4 %	0.1 %
factorization scale $t\bar{t}$	1.7 %	0.8 %	0.2 %	0.1 %
renormalization scale $W + \text{jets}$	2.9 %	3.5 %	2.8 %	3.5 %
final-state parton shower	1.1 %	2.8 %	0.4 %	2.0 %
QCDMistag	5.1 %	2.3 %	1.4 %	5.0 %
BJetMistag	7.0 %	4.3 %	3.4 %	7.6 %
total	16.4 %	11.5 %	8.8 %	15.5 %